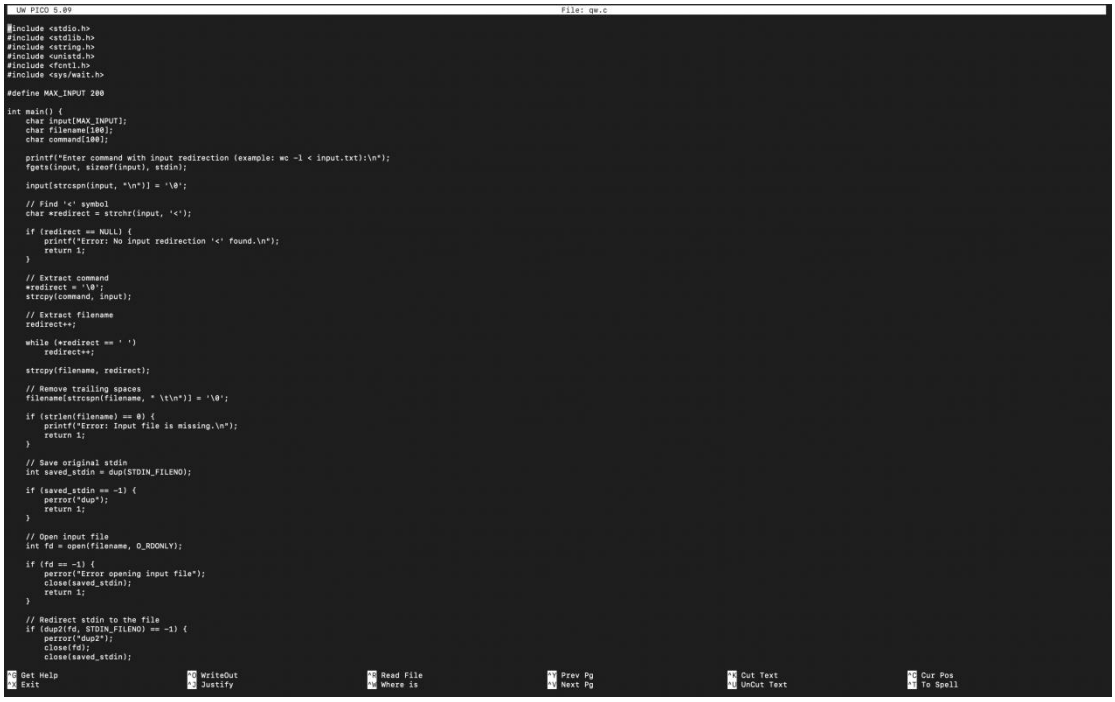
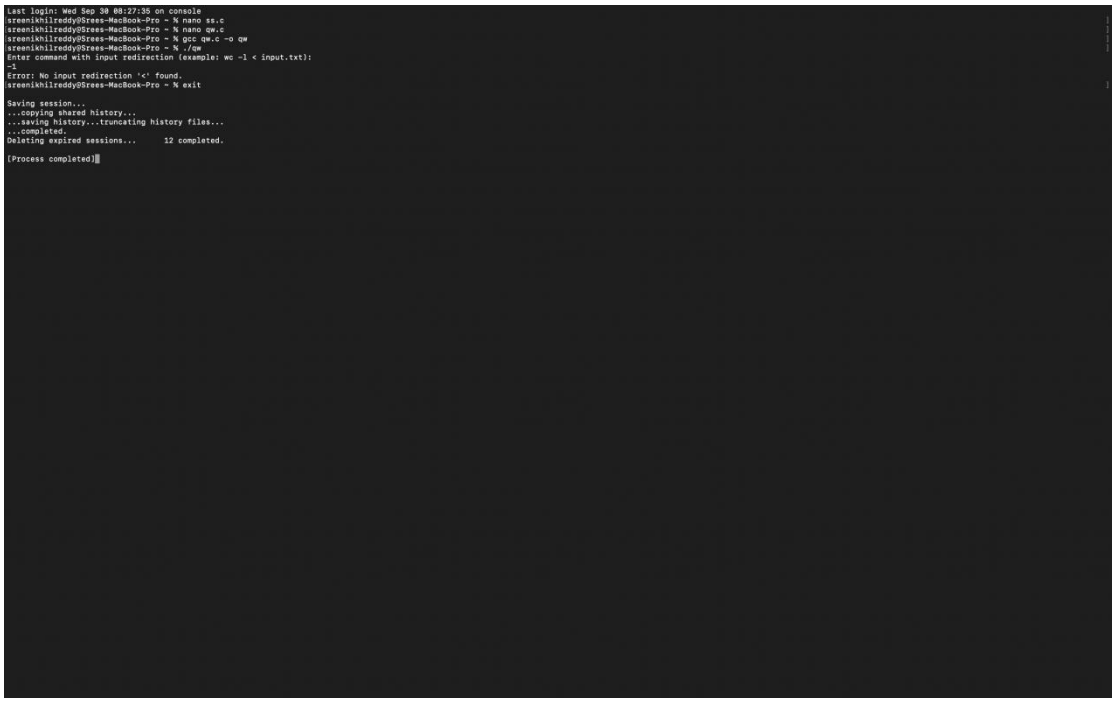
OSSP Skill Week -7

Roll No - 2520090058

To Parse Input Redirection, Open Files, Redirectstdin, Handle MissingFiles, Restore Streams,Test File Inputs



Output:



Task-2: To Parse OutputRedirection, Create OutputFiles, Redirect stdout,Handle Permissions, Verify Output Data, Test Redirection

```
 UW PICO 5.09                                   File: tt.c

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <fcntl.h>

#define MAX_INPUT 200

int main() {
    char input[MAX_INPUT];
    char command[100];
    char filename[100];

    printf("Enter command with output redirection:\n");
    printf("Example: echo Hello World > output.txt\n");

    fgets(input, sizeof(input), stdin);
    input[strcspn(input, "\n")] = '\0';

    // Find '>' symbol
    char *redirect = strchr(input, '>');

    if (redirect == NULL) {
        printf("Error: Output redirection '>' not found.\n");
        return 1;
    }

    // Separate command and filename
    *redirect = '\0';

    strcpy(command, input);

    redirect++;

    // Skip spaces
    while (*redirect == ' ' || *redirect == '\t') {
        redirect++;
    }

    strcpy(filename, redirect);

    // Remove trailing spaces
    filename[strcspn(filename, " \t\n")] = '\0';

    // Check whether filename is provided
    if (strlen(filename) == 0) {
        printf("Error: Output file is missing.\n");
        return 1;
    }

    printf("\nCommand: %s\n", command);
    printf("Output File: %s\n", filename);

    /*
     * Open/Create output file.
     *
     * O_WRONLY  -> write only
     * O_CREAT   -> create if file does not exist
     * O_TRUNC   -> erase old contents
     * 0644      -> file permissions
     */
    int fd = open(filename, O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd == -1) {
        perror("Error creating/opening output file");
        return 1;
    }

    // Save original stdout

^G Get Help    ^O WriteOut    ^R Read File    ^Y Prev Pg    ^K Cut Text    ^C Cur Pos
^X Exit        ^J Justify     ^W Where is     ^V Next Pg    ^U UnCut Text  ^T To Spell
```

Output:

```
sreenikhilreddy@Srees-MacBook-Pro ~ % nano tt.c
sreenikhilreddy@Srees-MacBook-Pro ~ % gcc tt.c -o tt
sreenikhilreddy@Srees-MacBook-Pro ~ % ./tt
Enter command with output redirection:
Example: echo Hello World > output.txt
hello world
Error: Output redirection '>' not found.
sreenikhilreddy@Srees-MacBook-Pro ~ % exit

Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.

[Process completed]
```

Task 3 -: To Parse Append Operator,Open Files in Append Mode, Redirect Streams,Preserve Existing Data, Validate Appends, Test Scenarios

```
 UW PICO 5.09                                File: hhh.c

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <fcntl.h>

#define MAX_INPUT 200

int main() {
    char input[MAX_INPUT];
    char command[100];
    char filename[100];

    printf("Enter command with append redirection:\n");
    printf("Example: echo NewData >> output.txt\n");

    fgets(input, sizeof(input), stdin);
    input[strcspn(input, "\n")] = '\0';

    // Find append operator >>
    char *redirect = strstr(input, ">>");

    if (redirect == NULL) {
        printf("Error: Append operator '>>' not found.\n");
        return 1;
    }

    // Separate command and filename
    *redirect = '\0';

    strcpy(command, input);

    // Move pointer after >>
    redirect += 2;

    // Skip spaces
    while (*redirect == ' ' || *redirect == '\t') {
        redirect++;
    }

    strcpy(filename, redirect);

    // Remove trailing spaces
    filename[strcspn(filename, " \t\n")] = '\0';

    // Check filename
    if (strlen(filename) == 0) {
        printf("Error: Output file is missing.\n");
        return 1;
    }

    printf("\nCommand: %s\n", command);
    printf("Output File: %s\n", filename);

    /*
     * Open file in append mode.
     *
     * O_WRONLY -> write only
     * O_CREAT  -> create file if it does not exist
     * O_APPEND -> append data at the end
     */
    int fd = open(filename, O_WRONLY | O_CREAT | O_APPEND, 0644);

    if (fd == -1) {
        perror("Error opening output file");
        return 1;
    }

    // Save original stdout

^G Get Help    ^O WriteOut    ^R Read File    ^Y Prev Pg    ^K Cut Text    ^C Cur Pos
^X Exit        ^J Justify     ^W Where is     ^V Next Pg    ^U UnCut Text  ^T To Spell
```

Output:

```
Last login: Wed Sep 30 08:39:57 on ttys000
sreenikhilreddy@Srees-MacBook-Pro ~ % nano hhh.c
sreenikhilreddy@Srees-MacBook-Pro ~ % gcc hhh.c -@ hhh
clang: error: unknown argument: '-@'
sreenikhilreddy@Srees-MacBook-Pro ~ % gcc hhh.c -o hhh
sreenikhilreddy@Srees-MacBook-Pro ~ % ./hhh
Enter command with append redirection:
Example: echo NewData >> output.txt
echo NewData
Error: Append operator '>>' not found.
sreenikhilreddy@Srees-MacBook-Pro ~ %
```

Task 4-: To Parsestderr Redirection, RedirectError Streams, CaptureError Output, VerifyFile Contents, Handle Failures, Test Error Cases

```
  UW PICO 5.09                                                File: hhhh.c

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <fcntl.h>

#define MAX_INPUT 200

int main() {
    char input[MAX_INPUT];
    char command[100];
    char filename[100];

    printf("Enter command with stderr redirection:\n");
    printf("Example: ls wrongfile 2> error.txt\n");

    fgets(input, sizeof(input), stdin);
    input[strcspn(input, "\n")] = '\0';

    // Find stderr redirection operator "2>"
    char *redirect = strstr(input, "2>");

    if (redirect == NULL) {
        printf("Error: stderr redirection operator '2>' not found.\n");
        return 1;
    }

    // Separate command from filename
    *redirect = '\0';

    strcpy(command, input);

    // Move pointer after "2>"
    redirect += 2;

    // Skip spaces
    while (*redirect == ' ' || *redirect == '\t') {
        redirect++;
    }

    strcpy(filename, redirect);

    // Remove trailing spaces
    filename[strcspn(filename, " \t\n")] = '\0';

    // Check filename
    if (strlen(filename) == 0) {
        printf("Error: Error output file is missing.\n");
        return 1;
    }

    printf("\nCommand: %s\n", command);
    printf("Error File: %s\n\n", filename);

    /*
     * Open/create error output file.
     *
     * O_WRONLY -> write only
     * O_CREAT  -> create if it doesn't exist
     * O_TRUNC  -> remove old contents
     */
    int fd = open(filename, O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd == -1) {
        perror("Error opening error file");
        return 1;
    }

    // Save original stderr

^G Get Help    ^O WriteOut    ^R Read File    ^Y Prev Pg    ^K Cut Text    ^C Cur Pos
^X Exit        ^J Justify     ^W Where is     ^V Next Pg    ^U UnCut Text  ^T To Spell
```

Output:

```
Last login: Wed Sep 30 08:40:34 on ttys000
sreenikhilreddy@Srees-MacBook-Pro ~ % nano hhhh.c

sreenikhilreddy@Srees-MacBook-Pro ~ % gcc hhhh.c -o hhhh
sreenikhilreddy@Srees-MacBook-Pro ~ % ./hhhh
Enter command with stderr redirection:
Example: ls wrongfile 2> error.txt
```